What We Teach About Race and Gender: Representation in Images and Text of Children's Books

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Motivation & Research Question

- Children's books teach kids about society and their place in it - in part through who is represented and who is absent.
- Persistent racial and gender inequality may be reinforced by the images and text children are exposed to early in life.
- Traditional content analysis is done by hand, which limits samples to a few dozen books and a few 'sites' (covers, main character).
- Research question: how are skin color, race, gender, and age represented in a century of influential children's books - and how has this changed over time?
- Approach: build and apply computer-vision and NLP tools to measure representation at scale, then examine the economic forces behind it.

Data: Influential Children's Books

- Books recognized by awards from the Association for Library Service to Children (from the 1920s on), split into two collections.
  - Mainstream collection: Newbery & Caldecott medals - honored for literary/artistic value, forming the 'canon' of children's literature.
  - Diversity collection: awards centering excluded identities (e.g., Coretta Scott King, Rise Feminist) - a plausible upper bound on representation.
- Mainstream books are checked out ~4x and sell ~5x as often as other children's books - evidence of outsized influence.
- Linked data: Seattle library checkouts, 1.5M+ Numerator book purchases with buyer demographics, Google Trends, U.S. Census, and CCES political attitudes.

Method: Turning Images and Text into Data

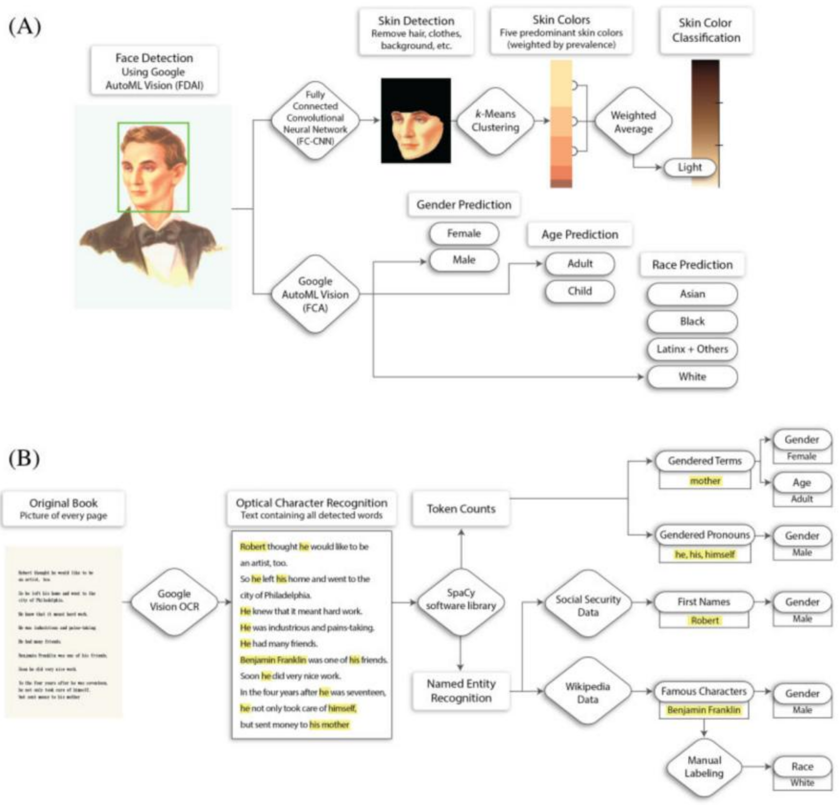

FIGURE II

Converting Images and Text into Data

Note: The two computational pipelines. Panel A (Image-to-Data): detect faces, segment skin, extract a representative skin color, and classify skin tint, race, gender, and age. Panel B (Text-to-Data): OCR the pages, then use token counts and Named Entity Recognition to measure gender, age, and race. (Figure II)

Image Analysis: Faces, Skin Color, Race/Gender/Age

- Challenge: 80%+ of images are illustrations, but off-the-shelf face detectors are trained only on photos.
  - Face detection: a custom transfer-learning model (Google AutoML Vision) trained on 5,403 hand-labeled illustrated faces; 93% precision, 77% recall.
  - Skin color: a neural-net segments skin pixels, k-means finds the dominant colors, converted to perceptually-linear L*a*b* space to score 'skin tint' 0-100.
  - Race, gender, age: a multi-label classifier trained on the UTKFace dataset (20,000+ labeled faces); 91% precision, 89% recall.
- Advantage over manual coding: rules (and any error) are applied consistently across every book and collection.
- Authors flag limits: binary gender, imperfect race prediction on illustrations, race as a socially-constructed category.

Text Analysis: Tokens, Named Entities, Names

- Pages are digitized with optical character recognition (OCR), then parsed with the spaCy NLP library.
  - Token counts: female vs male pronouns and gendered terms (e.g., 'queen', 'husband') measure gender; young vs old terms (e.g., 'girl' vs 'woman') measure age.
  - Named Entity Recognition (NER) identifies famous people, matched to Pantheon (70,000+ Wikipedia biographies) for gender; race hand-coded.
  - Character first names matched to Social Security name-frequency data to estimate the probability a character is female or male.
- Multiple independent measures let the authors cross-check gender findings and rule out artifacts of historical grammar conventions.

Result: Lighter Skin in Mainstream Books

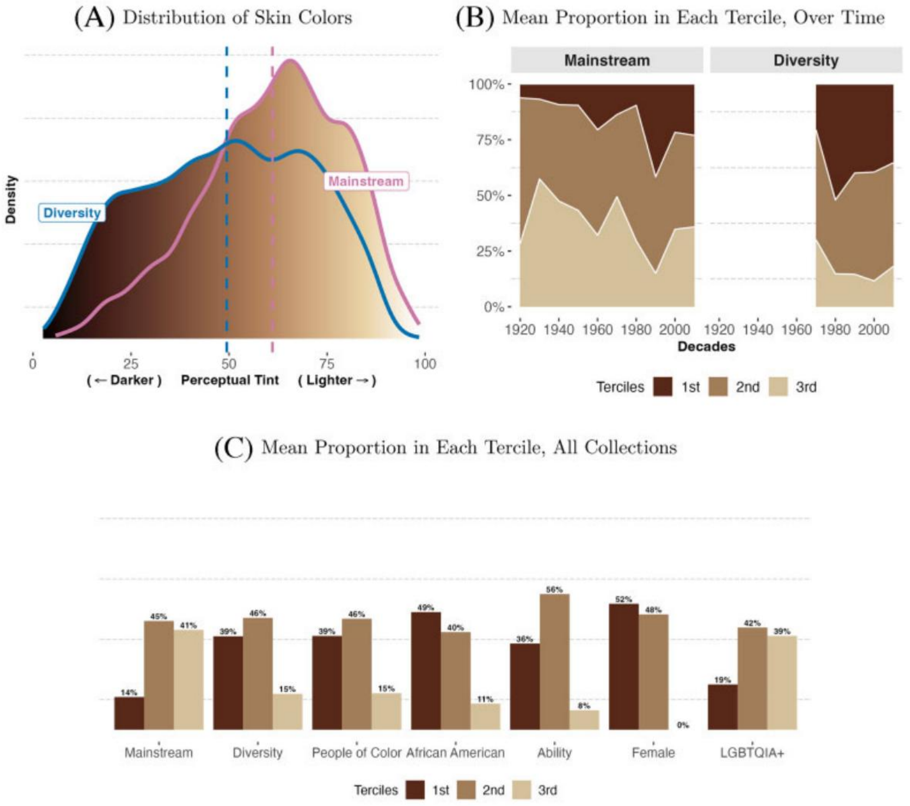

FIGURE IV

Skin Colors in Faces, by Collection: Human Skin Colors

Note: Across 44,000+ detected faces, Mainstream faces are lighter-skinned on average than Diversity faces (A; distributions statistically distinct, p<.001). Darker/medium terciles rise over time in both collections (B), but Mainstream change lags Diversity by decades. Mainstream has the lowest share of darker-skinned faces of all collections (C). (Figure IV)

Result: Black & Latinx Underrepresented vs Population

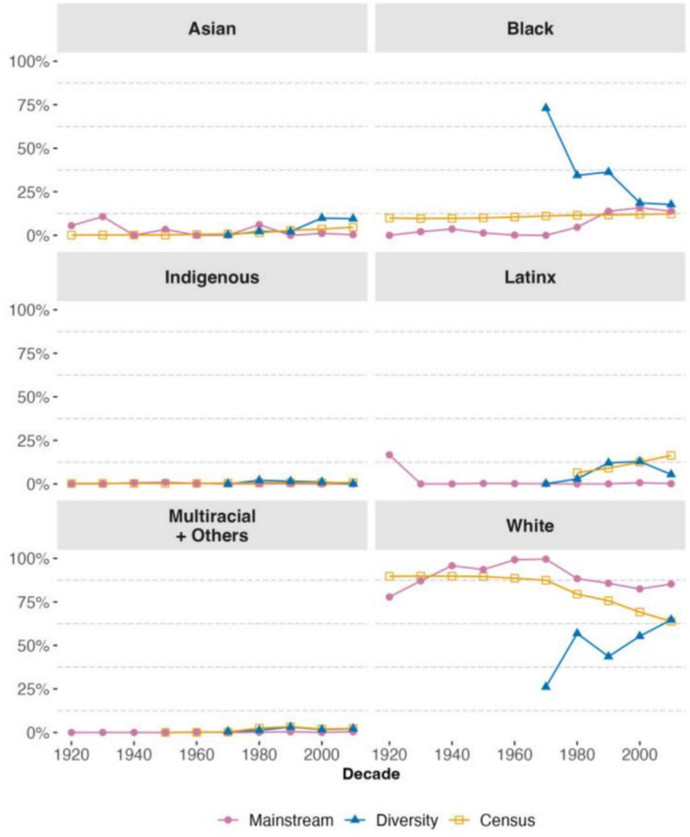

FIGURE VI

Share of U.S. Population and Famous People in the Text, by Race/Ethnicity

In this figure, we show the percent breakdown of famous people mentioned

Note: Share of famous people mentioned by race/ethnicity vs U.S. population share, by decade. In Mainstream books White figures are overrepresented while Black and Latinx figures are underrepresented; Black representation approaches parity only in recent decades. The average figure - pictured or famous - is White in every collection. (Figure VI)

Result: Females 'Seen but Not Heard'

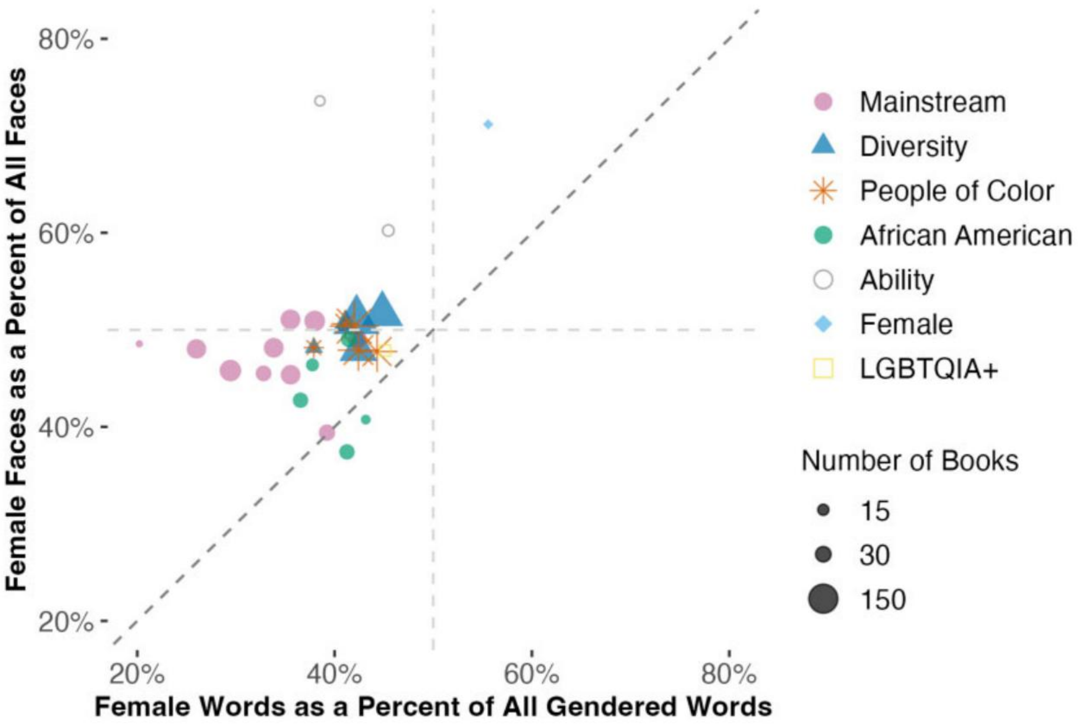

FIGURE IX

Female Representation in Images and Text of Children's Books

In this figure, we plot collection by decade average percentages of female rep

Note: Female representation in images vs text, by collection and decade. (Figure IX)

- Each point = a collection-decade: female share of faces (y) vs female share of gendered words (x).
- Most points sit above the 45-degree line: females appear more in images than in text.
- Interpreted as symbolic inclusion in pictures without substantive inclusion in the story.
- Females are under 50% in both images and text in nearly all collections; text share rises slowly over time but stays below population share.

Economic Forces & Contributions

- Demand side: people buy books that center their own identity - buyers with daughters choose more female content; book diversity correlates with local political views on race and immigration.
- Supply side: books centering nondominant identities are priced higher and are scarcer in libraries serving predominantly White communities.
- Contribution 1: new AI image-to-data tools, including a novel computational measure of skin color.
- Contribution 2: apply image + text tools to a century of influential books, documenting what changed and what endured.
- Contribution 3: link representation to economic forces that may transmit and perpetuate related beliefs across generations.
- Takeaway: computational content analysis makes large-scale, consistent measurement of representation feasible - and reveals persistent gaps.